

Generation Interconnection System Impact Study Report

For Queue Project Q15-0535

Woodsdale 500 kV
November 29, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0535. The Interconnection Customer proposes a 1057.6 MW natural gas combined cycle facility interconnecting at the Woodsdale 500 kV substation in the Ohio area. The request was received on November 10, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 283 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,059,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WESTWOOD GENERATION LLC to CORONET 500 KV line	MONTOUR STATION to ACADIA SUBSTATION 500 KV	104.3	2,945	9.4
SUNNYVALE to DRESDEN 230 KV	DRESDEN to CONESVILLE STA 500 KV	102.8	400	5.2
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500/345 kV Auto	HILLTOP 500/345 kV Transformer Bank	100.6	1,971	5.8
MINCO to BARRETT 230 KV	BARRETT to NORWICH 500 KV	100.3	538	11.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WESTWOOD GENERATION LLC to CORONET 500 KV line	Reconductor line	\$39,500,000	16% (\$6,475,000)
SUNNYVALE to DRESDEN 230 KV	Reconductor line	\$6,619,000	25% (\$1,684,000)
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500/345 kV Auto	Replace transformer with higher-rated unit	\$18,962,000	16% (\$3,109,000)
BARRETT to MINCO 230 KV	Reconductor line	\$6,926,000	55% (\$3,791,000)
	Total allocated to project		\$15,059,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.